$F_k = F_{k-1} + \alpha_k h_k$

GBDT ⇒ residuals ⇒ − gradient → loss fn → exponential → Ada boost

GBDT

x	y
	4
	6
	8
	12

loss ⇒ sq. loss

$\hat{y} = m_0$ (constant)

$(4-m_0)^2 + (6-m_0)^2 + (8-m_0)^2 + (12-m_0)^2$ → min

$$\frac{\partial}{\partial m_0} -2(4-m_0) + 2(6-m_0) + 2(8-m_0) + 2(12-m_0) = 0$$

$$4+6+8+12 - 4m_0 = 0$$

$$m_0 = \frac{4+6+8+12}{4} = \frac{30}{4}$$

$$F_{m-1} = m_0 + \alpha_1 h_1 + \alpha_2 h_2 \ldots \alpha_{m-1} h_{m-1}$$

$y \leftarrow$

$$F_m = F_{m-1} + \alpha_m h_m$$

→ x, y residuals

$$\frac{\partial \mathcal{L}(y, \hat{y})}{\partial \alpha_m} = 0$$

$F_m \Rightarrow F_{m-1} + \alpha_m h_m$

$$\frac{\partial \mathcal{L}(y, \hat{y})}{\partial \alpha_m}$$

→ sq loss

$$= \frac{(y - (F_{m-1} + \alpha_m h_m))^2}{\partial \alpha_m} = 0$$

$F_1 = m_0 + \alpha_1 m_1$

y → res → (x)

$$\frac{\partial \mathcal{L}(y, F_1)}{\partial \alpha_1} = 0$$

Random sampling

Base learner (n) (x) → n^1 d^1

stochastic GBDT

Bias – Variance

Depth of DT / base learner ⇒ [2, 5]

↑↑ overfitting

m / n – estimator

$$F_m(x) = m_0(x) + \sum_{i=1}^{m} \gamma_i h_i(x)$$

m ↑↑ overfit

m ↓↓ underfit

Regularisation by shrinkage

$$F_m(x) = h_0(x) + \sum_{i=1}^{m} \gamma_i h_i(x)$$

$$= F_{m-1}(x) + \nu \gamma_m h_m(x)$$

learning rate

$y = 10$

F_{m-1} 8 residual = 2

h_m $2 \times \nu$ 0.1

0.2 = 8.2

= 8 + 0.2 = 8.2

$\nu = 0.0000001$ (small) underfit

$\nu = 1$ (v large) overfit

$\nu \approx 0.1$

GBDT → m → depth → CS/DS

→ ν

Time ⇔ space complexity

Train time

1 DT ⇒ $O(n \log n \times d)$

n DT → $O(n \log n \times d \times n)$

RF → Parallel

GBDT ⇒ sequential

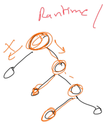
$\bigcirc_{m_0} \xrightarrow{\text{res}} \bigcirc_{m_1} \xrightarrow{\text{res}} \bigcirc_{m_2} \ldots$

Random sampling

$n \rightarrow n^1$

$n \gg n^1$

Runtime / Prediction



1 DT: $O(\text{depth})$

n DT → $O(\text{depth} \times n)$ ↓ small

$$F_m(x) = m_0 + \gamma_1 m_1 + \gamma_2 m_2 \ldots \gamma_m m_m$$

Outliers → model → parameters → loss fn

Depends

m ↑↑ ↑↑ highly affected

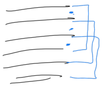
m ↓↓ ↓↓ less impact

Loss function

SE loss

absolute loss $|y - \hat{y}|$

RMSE



5 10 15 20 25

11 12 13 14 15 16 17 18 19